TERRACHAIN (LANDBLOCK)

Decentralized Land Registry & Title Transfer System
Detailed Technical & Architectural Report

Project Name	TerraChain — Decentralized Land Registry and Title Transfer System
Smart Contract Technology	Solidity v0.8.20, Hardhat Development Framework, OpenZeppelin ReentrancyGuard
Storage Architecture	Pinata Decentralized IPFS (InterPlanetary File System) REST API
Frontend & Web3 Integration	React 18, Vite, Tailwind CSS, Lucide Icons, Ethers.js v6 Context
Networks Supported	Localhost (Chain ID: 31337) & Ethereum Sepolia Testnet (Chain ID: 11155111)
System Status	Fully Implemented, 21/21 Unit Tests Passing, Verified Zero-Error Build

1. Problem Statement & Motivation

Land administration across the globe is plagued by double-selling fraud, paper deed forgery, corrupt database manipulation, and multi-year civil litigation backlogs. Centralized SQL databases introduce single points of failure where administrators or hackers can modify ownership records without public accountability.

TerraChain resolves this systemic problem by establishing an immutable, tamper-resistant registry on the Ethereum Virtual Machine (EVM). Real-world title deeds are anchored to decentralized IPFS storage, certified by government inspectors, and traded via atomic smart contract escrow.

2. System Architecture & 3-Tier Design

• **Tier 1: Smart Contract Layer**: Solidity 0.8.20 contract inheriting OpenZeppelin ReentrancyGuard, custom gas errors, and Checks-Effects-Interactions escrow.
• **Tier 2: Decentralized Storage Layer**: Pinata IPFS hosting surveyor blueprints and deed PDFs, returning immutable 32-byte CIDs.
• **Tier 3: Web3 Client Presentation Layer**: React 18 dApp using Vite, Tailwind CSS, Lucide icons, and an Ethers.js v6 provider/signer pipeline.

3. Smart Contract Implementation (LandRegistry.sol)

The contract stores land parcels with attributes: id, physicalAddress, areaSqFt, priceInWei, ipfsHash, currentOwner, isVerified, and isForSale.

Function	Access Control	Technical Description
registerLand(...)	Public / Any User	Registers unverified parcel on-chain.
verifyLand(id)	Admin Only	Approves title after cadastral inspection.
listForSale(id, price)	Owner Only	Lists verified parcel for sale at asking price.
delistLand(id)	Owner Only	Withdraws listed parcel from marketplace.
buyLand(id)	Payable + Guarded	Atomic purchase: pays seller, transfers title, refunds surplus.

4. Automated Unit Testing (21/21 Tests Passing)

All smart contract operations are covered by 21 automated unit tests in Hardhat and Chai. The test suite asserts deployment initialization, reverts on invalid input parameters, verifies admin title certification guards, confirms owner-only listing controls, enforces ReentrancyGuard, and mathematically verifies exact seller balance increases (+2.0 ETH) and surplus refunds.

5. Decentralized Document Storage (Pinata IPFS)

Storing full deed PDFs on-chain is cost-prohibitive. TerraChain stores documents off-chain on IPFS and registers only the cryptographic CID on-chain. The frontend includes DocumentUploader.jsx which offers drag-and-drop ingestion with client-side 15MB file size validation, live percentage progress, and instant gateway link generation.

6. Web3 Client & Dynamic Role Views

The frontend automatically computes the user's on-chain persona:
• **Admin / Inspector**: When connected wallet matches the contract deployer. Unlocks the Inspector Panel for title approval.
• **Verified Owner (<count>)**: When the wallet holds verified title entries. Enables listing/delisting controls in 'My Assets'.
• **Buyer**: Default persona for browsing marketplace listings and purchasing land via MetaMask.
• **Transaction Toasts**: Floating alerts providing real-time feedback for Pending, Success, and Reverted transactions.

7. Evaluator Live Demonstration Guide (3 Accounts)

1. **Start Services**: Local node on port 8545, deploy script, Vite server on port 3000.
2. **Seller Registers**: Connect Account #1, upload deed PDF to IPFS, sign registration transaction.
3. **Admin Verifies**: Switch to Account #0 (Admin tag appears). Open Inspector Panel, inspect deed, click 'Verify Title'.
4. **Seller Lists**: Switch back to Account #1 (Owner tag appears). Open 'My Assets', list for 2.5 ETH.
5. **Buyer Buys**: Switch to Account #2 (Buyer). Open Marketplace, click 'Buy Now'. Show seller receiving +2.5 ETH and title reassigned to buyer.

8. Oral Exam & Viva Defense Reference

Q1: How does blockchain prevent fraudulent physical land claims (The Oracle Problem)?

A: Blockchain guarantees ledger immutability, not physical truth. We resolve this via a two-tier gatekeeper model: entries start unverified and must be certified by a licensed Land Inspector. Judicial disputes can be flagged or delisted via contract functions.

Q2: Why store only IPFS CIDs on-chain instead of entire PDF deeds?

A: Storing 1 MB on Ethereum costs thousands in gas fees. Storing the 32-byte CID requires ~20,000 gas while providing cryptographic verification: altering even one character alters the CID, exposing tampering.

Q3: How is fund transfer secured against reentrancy in buyLand()?

A: Through OpenZeppelin's ReentrancyGuard and the Checks-Effects-Interactions pattern: state variables are updated before dispatching external ETH calls.

Q4: What happens if an owner loses their private key?

A: In production, Account Abstraction (ERC-4337) with Social Recovery allows designated trusted guardians (e.g., family, notary, registrar) to authorize key rotation upon legal probate proof.

Q5: Why choose an EVM architecture over Hyperledger Fabric?

A: Public EVM provides censorship resistance and zero-trust public transparency, preventing corrupt municipal actors from secretly altering private databases.